Decision Problems in Propositional/Predicate Logic

Learn the classifications for the exam!

Semantic Tableau Method

Decomposition Clues

→ conjunctive formulas

$$\begin{array}{c} A \land B \\ | \\ A \\ | \\ B \end{array} \qquad\qquad \begin{array}{c} \neg(A \land B) \equiv \neg A \land \neg B \\ | \\ \neg A \\ | \\ \neg B \end{array}$$

$$\begin{array}{c} \neg(A \to B) \equiv A \land \neg B \\ | \\ A \\ | \\ \neg B \end{array}$$

→ disjunctive formulas

$$\begin{array}{ccc} & A \lor B & \\ / & & \backslash \\ A & & B \end{array} \qquad\qquad \begin{array}{ccc} & \neg(A \land B) \equiv \neg A \lor \neg B & \\ / & & \backslash \\ \neg A & & \neg B \end{array}$$

$$\begin{array}{ccc} & A \to B \equiv \neg A \lor B & \\ / & & \backslash \\ \neg A & & B \end{array}$$

δ rules

$$\begin{array}{c} (\exists x)\, A(x) \\ | \\ A(c) \end{array} \qquad\qquad \begin{array}{c} \neg(\forall x)\, A(x) \\ | \\ \neg A(c) \end{array}$$

γ rules

$$\begin{array}{c} (\forall)x\, A(x) \\ | \\ A(c_1) \\ | \\ A(c_2) \\ | \\ \dots \end{array} \qquad\qquad \begin{array}{c} \neg(\exists x)\, A(x) \\ | \\ \neg A(c_1) \\ | \\ \neg A(c_2) \\ | \\ \dots \end{array}$$

$A(c_n)$ $\qquad\qquad$ $\neg A(c_n)$

$(\forall x) A(x)$ $\qquad\qquad$ $\neg(\exists x) A(x)$

δ before γ

<u>Ex 1.</u> Build a semantic tableau for $U = (q \wedge r \to p) \to (p \to r) \wedge q$ and decide the type of U.

$$U = \underbrace{(q \wedge r \to p)}_{\text{premise}} \to (p \to r) \wedge q \qquad (1)$$

$\neg$premise $\qquad$ β for (1)

$\neg(q \wedge r \to p)$ (2) $\qquad$ $(p \to r) \wedge q$ (3)

| α for (2) $\qquad$ | α for (3)

$q \wedge r$ $\qquad$ $p \to r$

$\neg p$ $\qquad$ q

| α for (4) $\qquad$ β for (5)

q $\qquad$ $\neg p$ $\quad$ r

r

open branch = consistency

$$DNF(U) = (\neg p \wedge q \wedge r) \vee (\neg p \wedge q) \vee (r \wedge q) \overset{\text{absorbtion}}{\equiv} (\neg p \wedge q) \vee (r \wedge q)$$

<u>Absorbtion laws</u> :

$$U \wedge (U \vee V) \equiv U$$
$$U \vee (U \wedge V) \equiv U$$

<u>Theorem of soundness and completeness</u>

logical conseq $\Leftrightarrow$ syntactic conseq.

Soundness : $U_1, \dots, U_{n-1}, U_n \vdash V \Rightarrow U_1, \dots, U_{n-1}, U_n \models V$

Completeness : $U_1, \dots, U_{n-1}, U_n \models V \Rightarrow U_1, \dots, U_{n-1}, U_n \vdash V$

Completeness: $U_1, \dots, U_{m-1}, U_m \vDash V \Rightarrow U_1, \dots, U_{m-1}, U_m \vdash V$

Distributivity of universal generalization over implication

$\neg U_1: \neg[(\forall x)(P(x) \to Q(x)) \to ((\forall x)P(x) \to (\forall x)Q(x))]$ (1)

| α for (1)

$(\forall x)(P(x) \to Q(x))$ (2)

|

$\neg((\forall x)P(x) \to (\forall x)Q(x))$ (3)

|

$(\forall x)P(x)$ (4)

|

$(\exists x)\neg Q(x) \equiv \neg(\forall x)Q(x)$ (5)

| δ for (5), c – new constant.

$\neg Q(c)$

| γ for (4), c – used for instantiation

$P(c)$

|

(4) copy:

| γ for (2), c – instatiation

$P(c) \to Q(c)$ (6)

|

(2) copy:

| β for (6)

$\neg P(c)$ ⊗ | $Q(c)$ ⊗